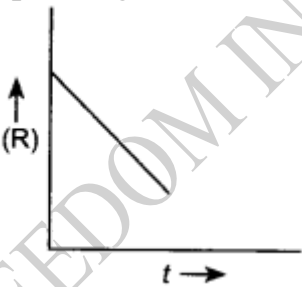


FREEDOM INTERNATIONAL SCHOOL
CHEMISTRY
L-3 CHEMICAL KINETICS
GRADE XII

1.	State a condition under which a bimolecular reaction is kinetically first order reaction.
2.	The rate constant of a reaction is $5.0 \times 10^{-5} \text{ L mol}^{-1} \text{ min}^{-1}$. What is the order of the reaction?
3.	Write the expression showing the change of concentration with time in the exponential form for a reaction of the first order.
4.	In the Arrhenius equation, what does the factor $e^{-E_a/RT}$ correspond to?
5.	Define the following terms: (i) collision frequency (ii) rate constant (k) (iii) activation energy (iv) half-life of a reaction (v) pseudo first order reaction (vi) order of a reaction (vii) molecularity of a reaction
6.	For a chemical reaction $R \rightarrow P$, the graph is plotted for the variation in the concentration, $\ln[R]$ vs. time (s). (i) Predict the order of the reaction. (ii) What is the slope of the line? (iii) Write the unit of the rate constant for this reaction.
7.	For a chemical reaction $R \rightarrow P$, the variation in the concentration (R) vs. time (s) plot is given as  (i) Predict the order of the reaction. (ii) What is the slope of the curve?
8.	For a reaction: $2\text{NH}_3(\text{g}) \xrightarrow{\text{Pt}} \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ Rate = k (i) Write the order and molecularity of this reaction. (ii) Write the unit of k.
9.	Derive integrated rate equation for the rate constant of a first order reaction.
10.	With the help of a labelled diagram explain the role of activated complex in a reaction.
11.	What is the physical significance of energy of activation ? Explain with a diagram.

12.	What is meant by the rate of a reaction? Differentiate between average rate and instantaneous rate of a reaction.
13.	How does change in temperature affect the rate of a reaction? How can this effect be represented quantitatively?
14.	Draw the plot of $\ln k$ vs $1/T$ for a chemical reaction. What does the intercept represent? What is the relation between slope and E_a ?
15.	What is the effect of adding a catalyst on (a) Activation energy (E_a), (b) Gibbs energy (ΔG) of a reaction?
16.	How does the order of reaction and its molecularity differ for a complex reaction ?
17.	For a reaction $2\text{H}_2\text{O}_2 \xrightarrow[\text{alkaline medium}]{\text{I}^-} 2\text{H}_2\text{O} + \text{O}_2$ the proposed mechanism is as given below : (1) $\text{H}_2\text{O}_2 + \text{I}^- \rightarrow \text{H}_2\text{O} + \text{IO}^-$ (slow) (2) $\text{H}_2\text{O}_2 + \text{IO}^- \rightarrow \text{H}_2\text{O} + \text{I}^- + \text{O}_2$ (fast) (i) Write rate law for the reaction. (ii) Write the overall order of reaction. (iii) Out of steps (1) and (2), which one is the rate determining step ?